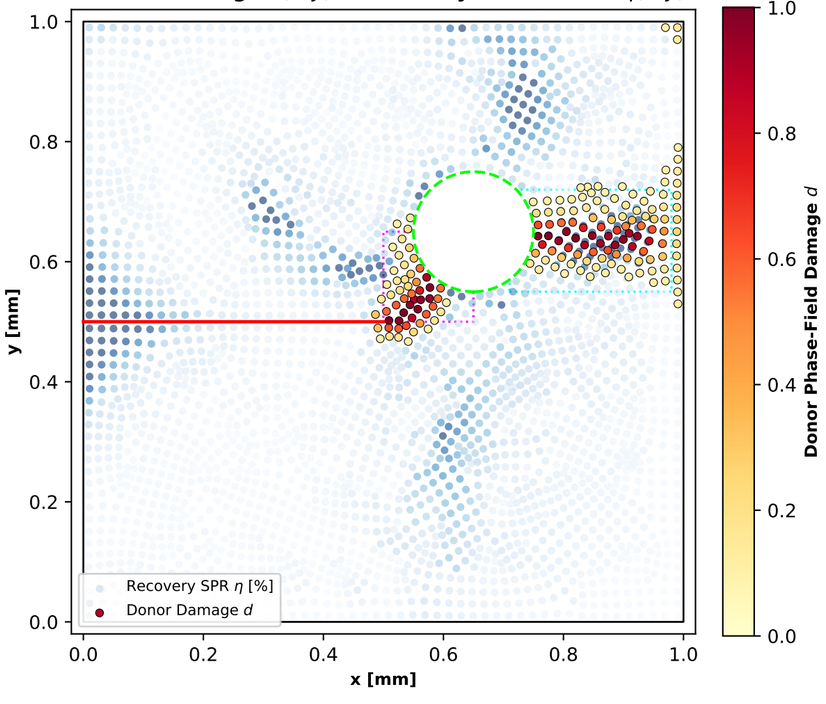
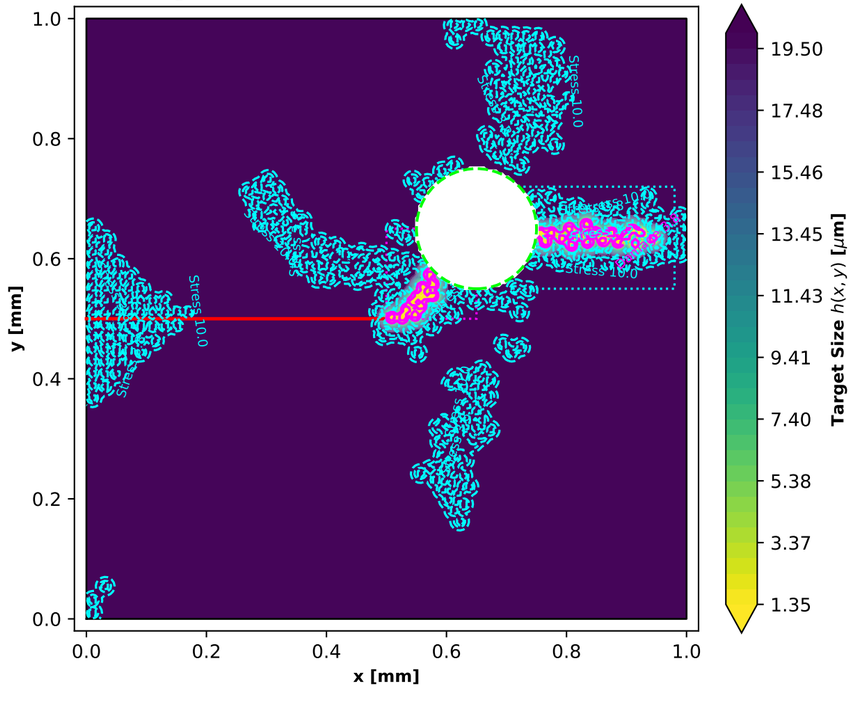
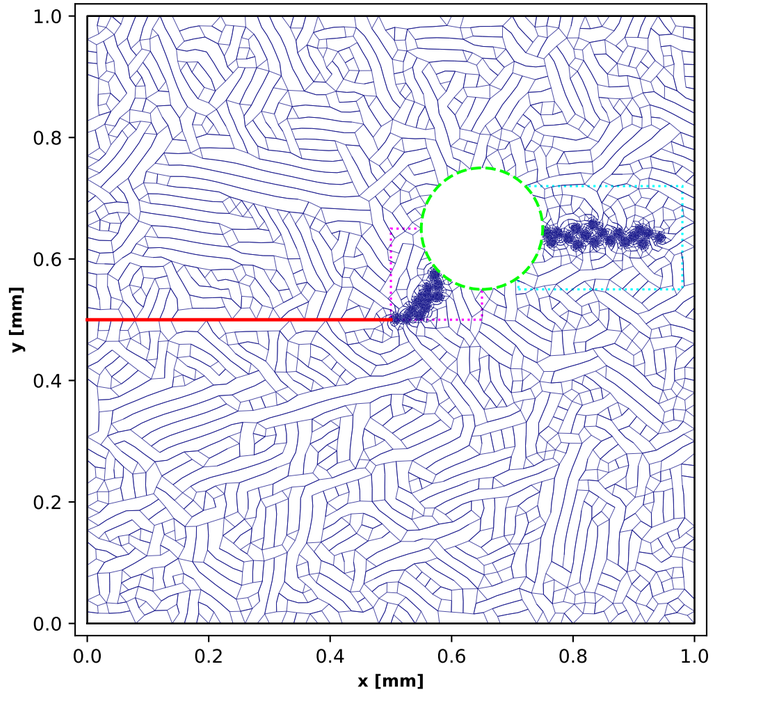
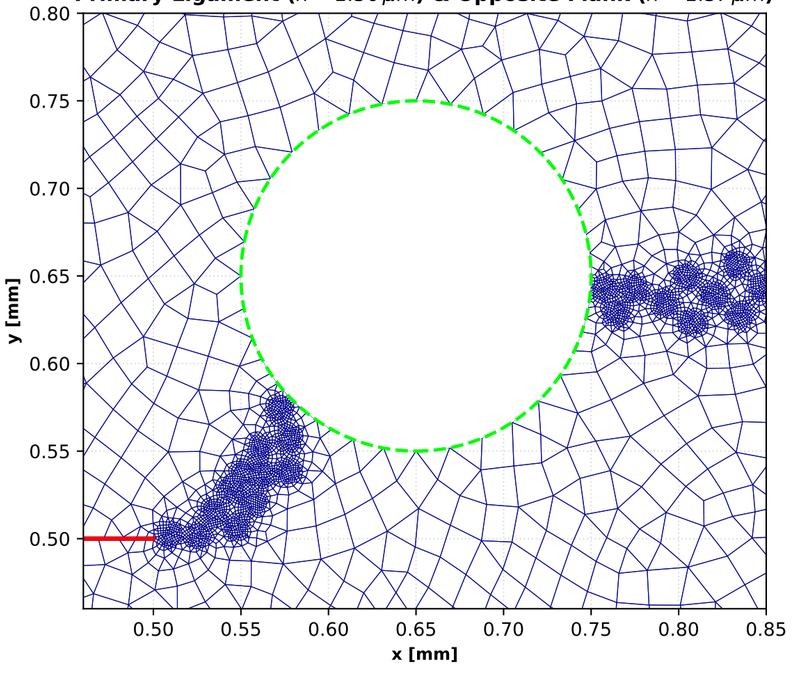
# TASK F1668: CONTROLLED THREE-WAY ADAPTIVE REMESHING STUDY

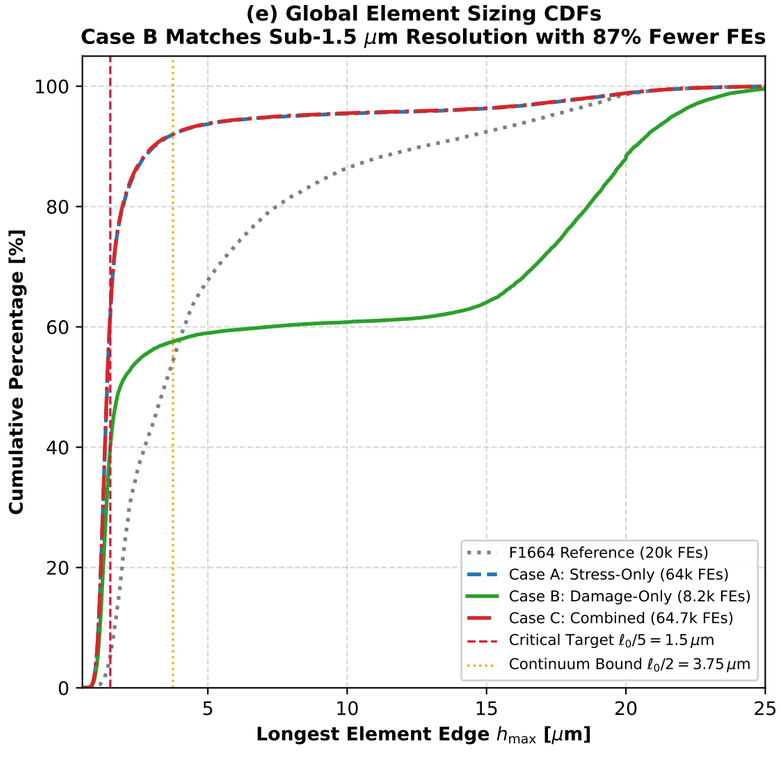
## Mode-I Plate with Off-Axis Hole | Authentic Coarse Donor & Recovery Fields | Identical Geometry & Mesher




**(f) Master Three-Way Scientific Evaluation Matrix**
**Case B Wins Efficiency-Resolution Trade-Off (8.2k vs 64.7k Quads)**

| Evaluation Metric | F1664 Ref | Case A (Stre | Case B (Dama | Case C (Hybrid |
|---|---|---|---|---|
| Total Elements | 20,213 | 64,032 | **8,237** | 64,746 |
| Pure Quads % | 97.18% | 78.03% | **77.67%** | 78.09% |
| Non-Pos Jacobians | 0 | 0 | **0** | 0 |
| Min Signed det(J) | 1.50e-7 | 1.10e-7 | **1.08e-7** | 1.29e-7 |
| Sub-1.5 um Count | 273 (1.4%) | 27,554 (43%) | **2,299 (28%)** | 28,097 (43%) |
| Ligament Median h | 4.684 um | 1.490 um | **1.501 um** | 1.491 um |
| Advancing Tip Med h | 2.091 um | 1.532 um | **1.570 um** | 1.516 um |
| Far-Field Median h | 5.530 um | 1.570 um | **20.609 um** | 1.573 um |
| Far-Field Bloat % | Moderate | Severe (72% | **MINIMAL (<1%** | Severe (72%) |
| Efficiency Score | Moderate | Poor | **EXCELLENT** | Good-Accurate |